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Stated Meeting Report

The Significance of Metals in Pre-Columbian Andean Culture

Heather Lechtman

The rich development of metallurgical technology that arose and was sustained in the New World prior to the Spanish invasion in the 16th century took place in the Andean zone of western South America in that area which is today Peru, Bolivia, Ecuador, and Colombia. Although Andean peoples supported a highly sophisticated metallurgical tradition with the production of a broad range of metals and metal alloys, little interest or attention has been paid to Andean metallurgy, perhaps because archaeologists and historians cannot boast of a "bronze age" or an "iron age" as characteristic of New World prehistory. Iron metallurgy was never developed in the Andes. Although both ancient varieties of bronze were invented there—the alloys of copper and arsenic and of copper and tin—and tin bronze was widely used and disseminated throughout its vast empire by the Inca dynasty, nevertheless these metals did not have the same impact on Andean civilization that they had among peoples of Europe and the ancient Near East.

If we were to ask the question, "What was the sphere of activities and interests from which metallurgy derived its greatest stimulation and achieved its most important developments among ancient societies of the New World and the Old?" we would come to see that the two metallurgical "revolutions" associated

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with bronze and iron in the Old World resulted from the demand for and impact of those metals primarily in two domains of life, warfare and transport, with agriculture running a close third. (In this discussion, the Old World covers a broad geographic area, including most of modern Europe and the Middle East, and a great depth of time, spanning the Bronze Age [ca. 3000-1200 B.C.] and Iron Age [ca. 1200 B.C.—A.D. 300].)

In Europe and in the Near East, both in the hand-to-hand combat of the foot soldier and in equestrian battle, the effectiveness of weapons was based largely on their piercing and cutting action. Knives, daggers, long and short swords, lance heads, spears, javelins, and battle-axes of bronze and later of iron became the new arsenal, replacing stone, wood, and bone weapons that could not compare in strength and durability. The manufacture of such offensive weapons was accompanied by the equally important production of their defensive counterparts, namely, body armor, some of the finest examples of which—such as the 7th-century-B.C. Cretan bronze helmets, corselets, and *mitrai* in the Norbert Schimmel collection—are often exhibited today in museums of art. Iron armor scales have been found that date to the 11th century B.C., accompanying the early use of iron for offensive weaponry in the Near East and the Aegean.

Although few examples remain of metal-rimmed wheels from chariots, wagons, carts, and other such vehicles, it is clear that the availability of bronze, and more especially of iron, for the manufacture of animal-drawn wheeled conveyances had a profound effect on long-distance travel and the movement of goods. In Europe during the period between 700 and 400 B.C., most weapons and tools of bronze disappeared and were replaced by iron. Nevertheless, iron was still an "expensive" metal. Many of the iron artifacts excavated from this period come from graves of the wealthy. Among these items are iron fittings of princely chariots: tires and nails, nave fittings, clamps, and wheel pegs. By the end of the 5th century B.C.,

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however, complex bridge bits and wheel pegs for chariots were much more common. By Roman times, the effectiveness of military legions in their movements throughout Europe in particular was dependent on ease of transport of the gear and provisions that accompanied them, much of it conveyed on wagons of wood and iron. Etruscan chariots, with bronze fittings and iron-rimmed, spoked wheels, gave way to Roman models, some of the most elegant of which—called by David Mitten the "Rolls Royces" of Roman chariots—have recently been found near the village of Šiškovci in Bulgaria, ancient Thrace, with dates in the late 3rd or early 4th century A.D.

The agricultural use of both bronze and iron was important in the Old World, although it was not primarily through the manufacture of metal tools for farming that the technology of metal production received its greatest impetus. The Early Iron Age (ca. 1200-900 B.C.) was a period of transition away from the use of bronze for all weapons and implements made of metal. We see a change from the exclusive use of bronze for plowshares, axes, adzes, and hoes at 12th-century-B.C. habitation sites in Cyprus and Palestine, for example, to the preferred use of iron for those same agricultural tools, as well as sickles, by the 10th century. Iron approached something akin to common use throughout the eastern Mediterranean, both for weapons and for tools, by the end of the 10th century B.C.

Turning to the Andes, we find that in neither the sphere of war nor that of transport did metals play an extraordinary role. There was no cavalry in South America prior to the introduction of the horse by the invading Spaniards. All combat was on foot. Hand-to-hand fighting involved the use of clubs of various kinds that depended on the crushing force of the blow delivered rather than on cutting or piercing. Of equal importance, however, were long-distance weapons that utilized missiles. Of these, spears and spear throwers, slings and shot were crucial to Andean styles of battle.

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It is of interest to explore why metals had such a small impact on Andean warfare, an otherwise obvious route for the development of metal technologies. What were the competitors of metals on the field of battle? It may come as a surprise that one of the chief competitors was cloth, a material used in both offense and defense.

Around 1615—roughly eighty years after the Spanish conquest of Tawantinsuyu, the Inca empire—Felipe Guamán Poma de Ayala, an Andean Indian, wrote a 1,200-page "letter" to King Philip III of Spain. His letter, *Nueva corónica y buen gobierno*, constitutes the first codex written in the Andes. Its purpose was to inform the Spanish king of the richness of Andean civilization as it was lived under the Inca and before the Inca and to decry the villainies of the Spaniards who had destroyed that great heritage. The letter is illustrated with some 397 line drawings Poma executed to accompany his text. Poma depicts the first and greatest of all the Inca emperors, Pachakuti Yupanqui (whose name meant "cataclysm"), in his role as soldier and as conqueror. The weapon that Poma chose above all to denote Pachakuti as warrior-emperor is the sling. Slings made by weaving, braiding, and plaiting animal and vegetable fibers were among the most important weapons used throughout Andean prehistory. They are, essentially, tools of cloth. Clearly, whether an adversary were hit by shot of stone or metal could not have made any great difference.

For body armor, soldiers wore quilted cotton tunics or else wound layers of cloth around their bodies. Most of the Spanish soldiers adopted quilted armor from the Inca, regarding it as superior to European steel breastplates, at least in the humid sierra. Inca soldiers hung round shields of hard chonta palm slats and cotton on their backs. Their heads were protected by quilted or wooden helmets. Instead of a shield made of wood and deerskin—such as the one Pachakuti holds in Poma's drawing—soldiers sometimes wrapped cloth around one arm to pad it against blows.

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In his other hand, Pachakuti holds the second most widely used Andean weapon, the club. Inca clubs such as his usually had pointed, star-shaped heads made of stone or bronze. Copper mace-heads of similar shape had been used much earlier, however—for example, those produced by the Mochica, a people who flourished along the north coast of Peru from about A.D. 0 to 600. Again, the effectiveness of metals as contrasted with stone in delivering a crushing blow was not of great significance.

Pottery was often a medium of sculpture through which ceramic craftsmen portrayed a wide range of activities of Mochica life. Often soldiers are shown, with their typical heavy clubs and padded or slat-and-cotton helmets. One of the most important and quite rare scenes of Mochica battle is painted along the flaring inner lip of a pottery vase currently in the collection of the Museum für Völkerkunde, Berlin. A group of Mochica warriors leads its captives, who are bound with ropes tied round their necks. The soldiers carry the typical Mochica club, but the important feature of this victory scene lies in the nakedness of the vanquished. Among Andean peoples, cloth was undoubtedly the item of greatest value, imbued with ritual significance, a symbol of rank, wealth, and power. It was not only used as a tool of war, both offensively and defensively, it also had a magico-military significance of its own, embodying the idea of strength and force. An enemy stripped of his garments was an enemy without retaliatory capacity. His force and his energy lay in his clothing.

Through ethnohistoric and archaeological study we have been able to reconstruct the panoply of weapons as they existed in the Andean highlands during Inca domination: the sling; the star-headed mace; the spear with tip of fire-hardened wood or of bronze; the long, sword-shaped double-edged club made of hard chonta palm wood; and the halberd with bronze head. Given the style of warfare in the Andes among both coastal and highland peoples, with weapons that depended on strength

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at impact rather than on a cutting edge, metal weapons did not confer great advantage either to the aggressor or to the defender.

Those who have traveled in the Andean area of South America—whether to visit Quito, Cuzco, or Machu Picchu—will have been impressed by the topography of that great land mass, a topography entirely dominated by the rugged and almost impassable Andes mountain chain. The Pacific coast of the central Andean zone consists of a narrow strip of barren, sandy desert from which the precipitous mountain slopes rise abruptly to the east. The wheel was neither invented nor could it have been utilized in this part of the world. Local fauna provided no draft animals or beasts of burden. The largest domesticated animals of the Andes are the camelids, the llama and alpaca. An adult llama typically carries only about 60 pounds. Movement, including the long distances traveled by the Inca armies, was entirely on foot along narrow roads, often hewn out of rock, which covered the entire length of the approximately 3,000-mile expanse of the empire. There was, in essence, no transport technology into which metals could be incorporated.

The combination of steep, vertical terrain and the unavailability of local fauna large and strong enough to be suitable as draft animals resulted in highland agricultural technologies based on terracing and, where terraces were not used, on long periods of fallow combined with the most conservative amounts of soil movements in tillage to minimize erosion and the ever present threat of downslope loss of soil by gravity. Thus the *chaqui taclla*, the Andean foot plow—a tool that functions as both hoe and digging stick—became the most important implement for planting, and in many highland areas it remains so. The blades of such tools were typically of stone, though bronze blades are known from the Peruvian north coast that date to about A.D. 1000, and modern blades are forged from iron. The animal-drawn plow was introduced by the Spaniards in the 16th century.

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In the Andes, metals performed in the realm of the symbolic, in both the secular and the religious spheres of life. They carried and displayed the content or message of status, wealth, and political power and reinforced the affective power of religious cult objects. We can recognize these functions in a series of artifacts ranging from the earliest consistent use of metal in the Andes to its employment during the Inca hegemony.

Status among Andean peoples, as in many societies, was instantly conveyed by what one wore in life and by what one wore at death. Earspools, for example, were the appurtenances of men of high rank—perhaps kings or priests. Some of the earliest and most splendid examples are of Mochica origin. Although we have spools of wood and others covered with feathers or inlaid with precious stones and shells, the finest are often made from metal. Their importance as conveyers of status and worldly or spiritual power was formally institutionalized by the Inca who alone were permitted to wear them. Poma's drawing of Pachakuti shows the emperor wearing such earspools. In Poma's illustrations, we can distinguish between members of the royal lineage and other individuals on the basis of their garments and whether they wear earspools. In fact, the Spaniards referred to the Inca as *orejones*—big ears—reflecting this practice.

Nose rings are another form of dress that was highly elaborated in the Andes, somewhat more among the peoples inhabiting what is today Ecuador and Colombia than among those in Peru. Nevertheless, recent finds along the far north coast of Peru, in the Piura River valley near the Ecuadorean border, have included some large, extremely fine nose rings of hammered silver and gold.

At death, it was the custom in many parts of the Andes to inter the deceased in the form of a mummy, the body wrapped around with cloth of a quality that denoted the individual's position within the community during his lifetime. In addition, those of highest status were sometimes also provided with masks of

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metal sewn onto or lying just beneath the outermost layers of cloth. The finest and usually the largest mummy masks are golden, though originally they were painted and decorated with plaques of copper and other materials. Those that are most often exhibited by museums and that appear in traveling shows, such as the recent "Gold of Peru" exhibit which toured the United States and Canada, are from the Chimú kingdom which dominated the north coast of present-day Peru between about A.D. 1000 and 1400.

The use of certain metals in the manufacture of religious objects as conveyers of spiritual power and for charismatic effect was no less important than their use in the secular realm. For example, during the Early Intermediate Period (ca. 200 B.C.—A.D. 600) on both the north and south coasts of Peru, a cult grew up around the practice of head taking during raids or wars on enemy groups. Archaeologists have called it the "trophy-head" cult, and its clearest manifestations occur among artifacts of the Paracas and Nasca cultures of south coastal Peru, from about 300 B.C. to A.D. 400. Large numbers of extremely rich graves have been excavated at the south coast site of Paracas. The mummy bundles, not all of which have been unwrapped, have yielded some of the finest cloth preserved from the prehistoric New World. Characteristic of these textiles are large, magnificent embroidered mantles that formed part of the mummy wrappings of high-status individuals of Paracas. A particularly fine mantle in the collection of the Museum of Fine Arts of Boston displays, in many-colored embroidery, masked and costumed cult figures in the form of birds, each holding a trophy head in one hand, two heads displayed on its chest, and two on its wings. The Ica River valley on the south Peruvian coast, not far from Paracas and Nasca, has yielded pairs of miniature gold trophy heads—each only 1 or 2 centimeters in height—that probably belong to a somewhat later Nasca version of related cult practice. They are compelling "charms"; their

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monumentality, despite their small size, is extraordinary.

* * *

All of the metal objects from the Andes described here as symbols of rank, power, and religious force are made of gold, silver, or a combination of those metals. Indeed, gold and silver had special ritual and political significance throughout Andean prehistory from their first important use as carriers of religious iconographic motifs associated with the Chavín cult that swept through the central Andes about 1000 B.C. up until their employment by the Inca royal dynasty as symbols of political power used solely by the emperor. All the vast mineral wealth of the Andean empire belonged to the Inca, and among the metals he controlled, gold and silver played a prominent role. They were his birthright, for the Inca dynasty began with the offspring of the sun and the moon. The first Inca was the son of the sun. In Inca cosmology, gold represented the sweat of the sun, silver the tears of the moon. Thus these two metals were intimately associated with the origin myth of the ruling dynasty and were second only to cloth as visible indexes of its wealth and power. Everyone is familiar with the stories recounted by invading Spaniards of Inca palaces whose walls were covered with sheets of silver and gold, with special rooms of miniature gardens that sprouted golden plants through whose leaves flew gold and silver birds. Of course, little of this wealth remains, for it was converted to bullion in the European melting pot.

Nevertheless, we do have a sufficient number of objects from all of the major periods of Andean prehistory to indicate a trend in Andean metallurgy that persisted throughout its development. That trend was motivated by a strong and consistent interest in the colors and quality of metallic gold and silver. It was important that metal objects have the appearance of gold and silver—their colors and their reflectivity—even if they incorporated very lit-

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tle of these precious metals in their structure. In fact, a large proportion of the gold- and silver-*looking* objects we have from the Andes are not made of the pure metals. Many contain relatively small amounts of gold or of silver.

The social arena in which metallurgy received its greatest stimulus in the Andes was the arena dominated by status and political display. An underlying cultural value system that appears to have strongly influenced the visual manifestation of status and power was a color symbolism oriented around the colors of silver and of gold. The most innovative and interesting aspects of Andean metallurgy arose from attempts by Andean metalsmiths to produce metallic gold and metallic silver surfaces on metal objects that were made of neither metal. These efforts resulted in the purposeful manufacture of binary and ternary alloys of copper, silver, and gold, and in a remarkable set of metallurgical and electrochemical procedures for gilding and silvering objects made of copper.

On the basis of the relatively small remaining corpus of metal objects of pre-Columbian date assigned to the Mochica culture, it has often been claimed, and rightly so, that the Moche peoples were among the most sophisticated of Andean metalworkers and that their products in metal were unequaled by those of the cultures that succeeded them, including the Inca. The discovery in the late 1960s of a large group of metal objects of Mochica style in the far north of Peru, at a site called Loma Negra near the Ecuadorean border, was of great importance, for it added substantially to the number of known objects either made or profoundly influenced by the Moche craftsmen. The majority of these objects are made of hammered sheet copper, individual, shaped pieces of which were often joined to produce a three-dimensional form. These artifacts are entirely covered with green corrosion products of copper that formed during burial, but originally their outside surfaces—and occasional-

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ly their inside surfaces also—were covered with extremely thin coatings of gold or silver. Such gilt or silvered surfaces can occasionally be seen when the green mineral layers are removed, revealing surface plating beneath. Thus, in their original condition, the objects appeared to be made of gold or silver.

Studies of the gold and silver surface coatings on the Loma Negra objects, carried out at MIT's Laboratory for Research on Archaeological Materials, proved difficult. The coatings are so thin, measuring 0.5-2 microns, that they often were not visible in cross sections at a magnification of 500. After many metallographic examinations of small samples removed from the objects, we came to three conclusions: (1) that the metal coatings on any one object are of a relatively uniform thickness and cover all surfaces, including the often paper-thin edges of the object; (2) that there is a solid-state diffusion zone between the gold and the copper indicating that, at some stage of the coating process, heat was applied; and (3) that there is no evidence of mercury gilding, the use of gold leaf or foil, or the flushing-on of molten gold on any of the objects.

The most impressive characteristic of these coatings is their extreme thinness and evenness. They look, indeed, very much like modern electrodeposits, which they could not possibly be. However, all of the features of the coatings could have been the result of some form of electrochemical replacement. After our experiments convinced us that the Loma Negra objects were not gilt by immersion in a bath of molten gold or gilt through any sweat welding or Sheffield plating type of process, and after mercury gilding was definitely ruled out, we attempted to reproduce the gross characteristics and the microstructures of these coatings using systems of electrochemical deposition that might have been available to ancient Andean metalworkers—that is, to deposit gold or silver onto copper without the use of modern chemicals, such as cyanide or aqua regia for dissolving the gold, and without the use of an external source of current.

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We have been successful in dissolving gold and silver in mixtures of corrosive minerals which were available to Andean metalworkers and which our earlier studies had shown were frequently used by them. The simplest and most effective method we have used for putting gold into solution consists of heating it gently for two to five days in an aqueous solution of equal parts of common salt (NaCl), potassium nitrate (KNO_3), and potassium aluminum sulfate ($KAl[SO_4]_2 \cdot 12H_2O$). This solution contains inter alia the same ions as are present in aqua regia to dissolve the gold in the form of trivalent ions. Chloroauric acid, $H(AuCl_4) \cdot 3H_2O$, would crystallize from the solution.

The resulting gold solution is highly acidic and immediately attacks copper dipped into it, causing a layer of copper corrosion products to form on the surface of the metal. It was thus necessary to neutralize the gold solution before introducing the copper. Among various salts that can be used for this purpose, we found sodium bicarbonate most effective. The optimum conditions for the gold in solution to plate onto the copper surface occur at a pH of 9. Copper sheet dipped into such a solution will be uniformly coated on all its surfaces (including the edges) with a film of gold approximately 1 micron thick after five minutes of gentle boiling.

We found that depositing a thin film of gold onto a copper substrate by simple electrochemical plating was only occasionally sufficient to bond the gold permanently to the copper. A more durable bond could be effected, however, by heating the gilt copper at a temperature high enough to produce solid-state diffusion of the gold and copper across their common interface. We proceeded to heat our gilt metal at temperatures ranging from about 500° C to 800° C for various periods of time. In the higher temperature ranges, excellent bonding occurred after only seconds of heating.

Most of the gold that has been analyzed from objects made on the north coast of what is today Peru, as well as native gold panned

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recently from rivers in that area, contains silver in amounts that range from a few percent to as much as about 15 percent by weight. We decided, therefore, to see if we could plate copper with alloys of gold and silver stemming from an aqueous salt solution containing various proportions of those two metals. We used copper pennies and successfully plated them by dipping them into a series of hot solutions/suspensions of silver and gold containing gold in concentrations of 100 – 5 percent by weight. We have not analyzed the actual plate to determine the silver:gold ratio on the surface of each penny, but it is clear that both gold and silver are deposited simultaneously and that the final color of the plate, which varies markedly from coin to coin, depends on the relative amounts of silver and gold in suspension and in solution at the moment of contact with the copper.

Ultimately, we were successful in achieving plates that were quite close in their visual characteristics and in their microstructures to the platings on the Loma Negra objects. The microstructures are typified by large, annealed grains within the copper sheet and, at the surface of the sheet, by heat-induced solid-state diffusion of the plated coating along the grain boundaries. Our copper sheet, plated by immersion with an 85 gold:15 silver coating, then annealed in a Bunsen flame for fifteen seconds, closely reproduced the Loma Negra microstructures and formed a well-bonded golden surface.

The dissolution of gold in aqueous solutions of corrosive salts is not a modern technique by any means. In fact, the gilding of iron and steel armor was achieved by such methods in 18th-century Europe and probably far earlier, as described by Godfrey Smith in his London edition of *The Laboratory or School of Arts* (1720). In these procedures, however, the original, aqueous gold solution was not used directly but was heated to dryness, and the resulting gold chloride ($AuCl_3$) dissolved from the complex mixture of salts by extraction with alcohol. It is the alcoholic solution that has

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traditionally been used to plate copper, iron, or steel with gold. By the 19th century, ether had replaced the alcohol for extraction.

Our method is much more direct. Through the simple expedient of neutralizing the aqueous gold solution with a common alkali such as sodium carbonate or bicarbonate or calcium carbonate, we can use the original solution, and no further extractions are required. Some such straightforward procedure is what we suspect the Andean metalworkers employed. Our previous studies had already demonstrated the probable use by these smiths of aqueous solutions of salt and niter or salt and ferric sulfate to gild objects by surface depletion methods. Such corrosive minerals were available to them and apparently were in common use.

When considering the plating mechanisms by which these incredibly thin and uniform layers of gold and silver were produced in the Andes, we recognize two categories of electrochemical plating without the use of an external battery or other source of electric current: "electroless" plating and "electrochemical replacement." In electroless plating, the solution itself contains its own reductant, and it will deposit metal on any surface that will catalyze the necessary electron exchange. In electroless plating, therefore, there is no chemical reaction with the basis metal—in our case, with the copper sheet.

By contrast, plating by electrochemical replacement occurs when a metal such as copper, high in the electromotive series (at the "negative" or "base" end of the series), is placed in an electrolyte containing ions of a metal, such as gold, lower in the series (at the "positive" or "noble" end). Chemically, a simple replacement reaction occurs, such as $2AuCl_3 + 3Cu \rightarrow 2Au + 3CuCl_2$. But such an equation does not reveal the mechanism, which is identical with that in the simple cells used by the first electroplaters. It is necessary to have anode and cathode areas, both of which have to be in contact with an electrolyte, and a complete circuit for electrons to flow from the

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anode through the metal to the cathode area, balanced by the return flow of ions through the electrolyte. Unless the geometry is such that anode areas are continually in contact with the electrolyte, there will be only an infinitesimal deposit before the electrode potential is everywhere the same and the action stops.

In replacement plating, different parts of the same metal surface provide both the anode and the cathode. In our case, as in the case of the Loma Negra objects, we assume that small pits or irregularities on the surface of the copper sheet initially act as anodes and continue their anodic activity until they are completely blocked or protected from the electrolyte by the deposit—the gold—which plates onto adjacent cathodic surfaces. The anodic and cathodic areas will be in delicate balance, depending on the details of ion availability and polarization. A good deposit occurs only when the anode areas shrink to the point of near invisibility, but they must not become completely sealed off from the electrolyte. Plating formed by replacement mechanisms should therefore be characterized by small anodic areas surrounded by much larger cathodic ones.

Mochica metalsmiths were not limited solely to gilding and silvering objects made of copper. Their desire to achieve culturally valued color effects was played out in the alloy systems they developed or invented, some of which have come to be considered the hallmark of Andean prehistoric metallurgy.

The alloy systems developed during the Early Intermediate Period are extremely important, for they continued to be used in later periods, and some of them dominated the Andean metallurgical scene up to the time of the Spanish invasion. The earliest of these that has been identified archaeologically is the alloy of copper and silver which the Mochica used over a wide range of silver concentration, from a few percent to over 30 percent, by weight, of silver. The alloy became particularly popular both on the north coast, within Chimú territory, and on the south coast, where it was used

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extensively by Chincha metalworkers during the Late Intermediate Period.

Copper-silver alloys have two properties that were important for Andean craftsmen: their toughness when hammered and their development of enriched silver surfaces when hammered and annealed. These alloys were used almost exclusively for the manufacture of objects made of sheet metal. Even with silver concentrations as low as about 5 percent by weight, the metal becomes hard but not brittle when hammered into thin sheet. The flexibility and toughness of copper-silver sheet metal allowed it to be shaped easily and, once formed, to retain its shape with far greater strength than pure silver or even sterling silver (7.5 percent copper). It was thus an excellent material for metalworkers whose forte was the production of items from elaborately hammered and joined pieces of metal sheet.

The surface-enrichment properties of copper-silver alloys are well understood. Particularly for alloys containing about 10 percent or more of silver by weight, the repeated sequences of hammering, annealing, and removing the surface copper oxide scale formed on annealing—sequences necessary to the fabrication of sheet metal from the alloy ingot—produce enriched silver surfaces on the resulting sheet as the surface copper is removed. Thus metal made from such alloys, of mottled copper color when cast, is bright silver in color after having been hammered into sheet. The formation of silvered surfaces on objects hammered from these alloys is an inescapable consequence of annealing and of the attendant loss of surface copper through oxidation. There is, essentially, no way of preventing it.

Copper-silver alloys were used by the Mochica to produce objects of sheet metal because the sheet was hard and tough and because objects made of such sheet looked like silver. In later periods, these alloys continued to be used for the same reasons, and the well-known *vasos retratos*, or effigy beakers, of Chimú and Chincha origin, said to be of silver, are

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sometimes made of copper-silver alloy.

By far the most important alloy system developed during the Early Intermediate Period and often used by the Mochica in the manufacture of sheet metal objects was that of copper-gold. Copper and gold when melted together form a complete solid solution series throughout the entire range of possible alloy compositions, and objects varying widely in composition have been encountered. Silver is also often found in these Andean alloys either because the gold used contained some silver, as placer gold from the Andes often does, or because gold was added to a copper-silver alloy.

Copper-gold alloys, like copper-silver alloys, become hard on hammering but retain their flexibility. They were, therefore, perfectly suited to the sheet metal tradition already characteristic of north Peruvian metalworking. But these alloys were used and subsequently highly developed primarily for another property—the gold color that they confer on articles made from them once the surfaces of such objects are suitably treated. For example, the Mochica ingot containing 60 percent copper and 31 percent gold is not golden but distinctly copper in color in its cast condition. As in the case of the copper-silver alloys, when such an ingot is hammered to produce thin metal sheet, copper is lost from the surfaces of the alloy through oxidation on annealing. Objects made from copper-rich *tumbagas* of this type soon develop deep golden surfaces as increasing amounts of surface copper are lost in the hammering and annealing process. When silver is also present, the surfaces of the object may require additional chemical treatment to remove some of the silver as well, thereby enriching the gold. We now know, in fact, that Andean metalworkers were capable of dissolving silver from such enriched gold-silver surface alloys with naturally occurring acid minerals, a highly sophisticated chemical process. Evidently this discovery had already been made by the Mochica, though it was most widely adopted by the Chimú who, several centuries later, dominated the entire north coast of Peru.

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There is no question but that Chimú metallurgy was a continuation of and highly dependent on that of their predecessors, the Mochica. Chimú metalsmiths took advantage of Mochica know-how but used it in a slightly different way. Many of the large and lushly golden mummy masks from Peru familiar to museum goers are of Chimú origin. The largest known example of these, presently in the collection of the Metropolitan Museum of Art in New York City, measures 29 inches wide by 16 inches high. Its golden surface is dazzling, yet the mask is made of a ternary alloy containing only 39 percent gold, the remainder being silver (49 percent) and copper (12 percent).

As in the case of binary silver-copper alloys, when ingots of the ternary copper-silver-gold alloy are forged into metal sheet, a point is reached when continued hammering to reduce the thickness of the ingot causes the metal to become brittle with the possibility of fracture if the hammering proceeds too far. When the metal reaches this stage, the smith anneals it. The heavily worked metal grains recrystallize and regain their ability to be plastically deformed. Hammering can then proceed until, once again, the metal becomes highly stressed and brittle. Annealing serves to relax the crystalline structure so that forging can proceed. At each anneal, however, some of the copper atoms in the near vicinity of the metal surface move to the surface under the action of the heat and oxidize there, forming a brown to black scale that covers the surface. This scale must be removed before hammering can continue. Andean smiths could have used certain acid plant juices or stale urine to pickle off the copper oxide scale quite easily. After many sequences of hammering, annealing, and pickling, the resulting thin metal sheet will have lost so much surface copper that the surface becomes enriched in silver and gold through copper depletion. A mummy mask at this stage of manufacture would appear as if made of silver, since the binary silver-gold alloy remaining at the surface is silvery white in color.

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But the mummy masks are brightly golden. Chimú smiths, approaching the task of gilding from the same point of view as that of silvering, were faced with the problem of removing the silver in the surface silver-gold binary alloy in order to leave only the gold in situ. A modern chemist might use various cyanide solutions or distilled nitric acid to part silver from gold, but we can assume that such distilled acids were not available to Andean smiths. What they did have at hand and apparently used, however, were combinations of naturally occurring acid minerals—such as ferric sulfate and sodium chloride (common table salt)—which, in an aqueous environment, effectively remove silver from a silver-gold alloy, leaving the gold in place.

This set of procedures—which removes first copper, then silver from the surface of a ternary alloy containing copper, silver, and gold, thereby gilding the surface—is known as depletion gilding and accounts for the configuration of the electron microprobe concentration profiles of the cross section removed from the Metropolitan's mummy mask. We cast an ingot with precisely the same ternary composition as the Metropolitan's mask and hammered it into thin sheet through many rounds of annealing and pickling. When we were through, our sheet looked like silver. We covered the sheet with aqueous pastes of ferric sulfate and salt or ferric sulfate, salt, and iron oxide. These were allowed to remain at room temperature for two days, when they were removed and washed. The resulting sheet had a reddish-brown color, the color of parted gold in finely divided, particulate form. But when burnished or gently heated at about 300° C for half an hour, the gold immediately became a rich, smooth, and shiny yellow color. Nothing could be simpler!

Depletion gilding, which relies on the removal from the surface of an alloy of its baser metal constituents in order to leave the noblest metal in place, was used effectively by the Chimú to gild sheet metal objects that con-

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tained as little as 12 percent gold by weight, the remainder of the alloy being principally copper with some silver.

The *tumbaga* alloys with their inherent gold enrichment properties swept through the Americas from Peru to Mexico and were in common use in that entire region when the Spaniards invaded Central and South America in the 16th century. They constitute the most significant contribution of the New World to the repertoire of alloy systems developed among ancient societies. In the Central Andes, copper-rich *tumbagas* continued to be used after the Early Intermediate Period primarily to produce gold-colored objects of sheet metal in contrast to their use in Colombia, the Isthmian area, and Mexico, where they were employed primarily in castings.

* * *

There has been considerable discussion about why *tumbaga* had the impact it had, why it spread so far and overcame the particular metalworking traditions of the various cultures that adopted it (e.g., the almost diametrically opposed traditions of Peruvian and Colombian artisans, the former exquisite forgers of metal, the latter superb founders), each tailoring the concept of depletion and enrichment to its own techniques. Explanations have invoked the economy of the system—that is, one can spread one's gold much farther if objects are made of *tumbaga* rather than of pure gold. But if one argues for the economizing of gold, one has also to bear in mind that in alloys of *tumbaga* all the gold inside the alloy is "wasted." Only that at the surface is functional in the sense that it is visible. Using gold leaf or other external plating systems would be much more economical. In fact, metal objects were gilt with gold foil, and William Root examined a number of objects made of copper-gold *tumbaga* from Coclé, in Panama, and from Costa Rica, Guatemala, and Yucatán in which the gilding was achieved by applying the foil to the surfaces of the *tumbaga* castings! Clearly in

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these cases the gold within the alloy was present for a purpose entirely different from any possible gilding function. The ternary copper-silver-gold *tumbagas* were also used, at least in parts of Colombia, in the manufacture of tools such as awls and axes which were cast and selectively work hardened to rival the best Inca products in copper-tin bronze. Thus it was a serviceable alloy for tool production, though it was used much less frequently for making tools than for almost every other variety of object produced by the Colombians—nose rings, *tunjos*, vessels, ornaments, and so forth. Root argued that although *tumbaga* might have been used because it made casting easier, it made objects harder, and it made a small amount of gold go farther, the most likely reason for its employ was that the people who made and used it preferred the color of gold to that of copper—or, to put it another way, they liked the color of *tumbaga* more than that of gold *or* of copper.

While all these explanations certainly enter into the constellation of *tumbaga*, there is another consideration that lies more in the realm of attitudes: attitudes of artisans toward the materials they used and attitudes of a culture area toward the nature of the technological events themselves. We are really seeking explanations, not for the use of the particular copper-gold alloy called *tumbaga*, but for the development, geographical spread, and persistence for over two millenia of systems of surface enrichment that stimulated the invention of a variety of alloys and that were adaptive to quite disparate traditions of handling metals.

The basis of Andean enrichment systems is the incorporation of the essential ingredient—the gold or the silver—into the very body of the object. The essence of the object, that which appears superficially to be true of it, must also be inside it. In fact, the object is not that object unless it contains within it the essential quality, even if the essence is only minimally present. For, without the incorporation of the essence, its visual manifestation

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is impossible. Although these enrichment systems—whether of silver or gold—have been used by metalworkers in other areas of the world, the role they played in the Andes is unique. Almost from the earliest appearance of metallurgy there, depletion and enrichment processes assumed a special place that persisted throughout the entire course of Andean metallurgical development and were responsible for stimulating some of its most interesting achievements. Although ideological considerations may have had little to do with the initial working out of these procedures, it seems certain that the way in which Andean peoples perceived such processes—or at least the objects that resulted from their use—had a great deal to do with the way in which the technology emerged and matured. Belief systems and attitudes toward materials supported the technology and gave rise to further developments along similar lines.

On another occasion, I suggested that one of the ways to test the hypothesis concerning incorporation of essences as an ideological motif underlying the elaboration of Andean metal gilding technologies and its possible, more widespread occurrence behind a style of technological behavior within Andean society would be to examine those technologies that surround what is unequivocally the most important arena of Andean material culture—cloth production. Do similar concerns appear to have operated in the elaborate weaving techniques utilized by Andean specialists in the prehistoric period?

No other item of manufacture in the Andes assumed the importance of cloth. By the Incaic period, cloth and agriculture were considered the dual forms of wealth, and textile manufacture had behind it some 4,000 years of development. In Andean society, cloth performed in virtually all aspects of life—it had religious and ceremonial functions, political and military functions; it was the commodity that carried with it the highest status and, for the Inca, was the item most valued by the state and the individual alike. Textiles were the

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primary visual medium for the expression of ideas, the fundamental art form of Andean peoples. And the manipulation of fibers—whether for the thatching of roofs, the building of bridges, or the weaving of garments—formed one of the primary systems of Andean technology.

Although from quite early periods of Andean prehistory it is clear that groups of specialists were involved in the production of elite textiles, it is equally evident that virtually everyone of all ages spun and all women and some men wove. The technology was a pervasive, a universal, element of Andean life, and this intense involvement, in conjunction with the social-ideational significance of cloth, gave rise to the elaboration of an extremely wide range of highly sophisticated weaving techniques. All scholars agree that Andean textiles rank among the finest of preindustrial cloth manufactures.

One of the most impressive aspects of Andean weaving arises from the preponderant use of "structural" as opposed to "suprastructural" techniques for realizing the patterns carried by the finished, woven cloth. When a design or pattern is achieved through suprastructural means, the design is added to a completed web. Its removal from that web would in no way affect the integrity of the woven cloth which acts as a support for the design but is otherwise structurally unaffected by it. Embroidered designs fall in this category, as do brocading and painted cloth; dyed textiles, achieved by tie-dyeing or, in a sense, ikat, are also included in this group. Such techniques were used by Andean weavers, particularly by societies that developed along the Pacific coast of South America. In contrast, designs achieved by structural means are realized through the manipulation of warp and weft yarns during the process of weaving so that the design emerges from the structural rendering of the web. To remove the design would be to destroy—literally to unweave—the cloth. Tapestry techniques; double, triple, quadru-

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ple cloth; gauze weaves, and warp patterned weaves as well as plain weaves are only some of the Andean cloth types whose characteristics are achieved through manipulation of the structural elements of the fabric. There is no question but the primary creative direction of Andean weaving—both synchronic (area to area) and diachronic (over time)—lay in the rich elaboration of structural techniques, in the playing out of a structural technological style. As Anna Gayton observed when discussing ancient Peruvian textiles, "This potency of structure, sufficient to enhance or mar the qualities of yarns, colors, and designs, was a major motivation for developing the different methods of interlacing warps and wefts which distinguish one weaving technique from another."

When we compare Andean weaving with any of the other great traditions of cloth manufacture, whether Oriental or European, nowhere else do we find such an elaboration of and commitment to a broad range of weaving techniques based on loom yarn manipulation. It is not simply that Andean cloth production greatly emphasized and relied on patterns structurally rendered but that the complexity of such structural manipulations—as in quadruple cloth—went far beyond mere virtuosity. William Conklin argues persuasively that the reason behind such a rich field of structural invention in Andean weaving lies in the role of textiles as the chief carrier of cultural message in conjunction with the availability of the technology at all levels of the society.

What we must ask is whether the visual message carried by Andean cloth, to the extent that that message was borne by the design motifs, had to be contained in and to be generated by the very structure of the fabric itself. That is, just as in the case of the depletion gilding of metal, does there come a point at which the technology per se becomes the medium for the expression of message? In the case of Andean highland weaving, intricate manipulations of many warp planes and myriad weft yarns were, strictly speaking, unnecessary to the accom-

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plishment of the designs. Simpler techniques could have achieved the same patterns. But Andean weaving seems to have responded to notions that saw the achievement of visual, surface message as emerging from underlying, invisible structural relations. Andean weaving insists that message be embodied in and expressed by structure.

Although this treatment of Andean cloth techniques is perforce summary, we do see that key technologies associated with cloth and metal production shared stylistic modes, perhaps because those modes are expressions of cultural ideals, that is, they incorporate ideological concerns of the society at large. Lexicographic and ethnographic studies of Andean metallurgical and weaving vocabularies should certainly help to bridge the realm of technical process with that of cosmic ideas about universal processes, and ethnohistoric research on craft production and on Andean cosmologies is absolutely essential in this regard. For example, both Regina Harrison and Gerald Taylor, ethnohistorians of the Andes, in recent studies of late 16th- and early 17th-century Quechua religious texts, call attention to the use in these texts of the Quechua term *camay*, the act of infusing life spirit into an inanimate object. In various ritual poems that refer to the acts of creation by Viracocha, the creator, the term *camay* is used expressly to denote Viracocha's animating or breathing a spirit into an object. *Camay* refers to the domain of the material and of the concrete, to the domain of people and of the natural and cultural objects they fashion and use. Perhaps the notions of "technological essence"—of the visually apprehended aspect of an object as revealing its inner structure—are related to these fundamental Andean concepts of the divine animation of all material things.

In a recent and provocative article about the relations between precious metals and politics in the pre-Columbian era. Mary Helms compares those sets of relations as they were manifest in the Andean area, particularly in Incaic Peru, and in the Intermediate Area of

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lower Central America and northern South America (including portions of Nicaragua and Honduras, Costa Rica, Panama, Colombia, and northern Ecuador). For each of these two areas she describes the overriding metallurgical-technological style, the political organization, and the ideologies of the ruling elites of the societies in question, seeking interpretations that correlate metallurgical traditions with the role of political elites. Her discussion of Incaic metalworking includes not only its bias toward gold or gold-appearance but its commitment to a long-standing Andean tradition of handling metal, a tradition based on forging and joining pieces of metal to achieve three-dimensional forms, as contrasted with the tradition of casting metal by the lost-wax method which was overwhelmingly the practice in the Intermediate Area. The Andean style she calls "architectural" in the sense that objects were built or constructed from thin metal sheets. For the ruling Inca elite, such constructions were often of gold or gold-appearing metal:

> In short, clothing, buildings, utensils, even (as effigies) people and animals—all fundamental elements of the cultural setting in which the nobility lived either symbolically or, to the extent allowed by actual use of these golden goods, behaviorally—could be built from sheets of gold and silver. . . .
>
> A simple and obvious message seems to be communicated by such constructions. The Inca nobility apparently wished to be construed as living in a world *inherently* composed of the qualities of the celestial realm from which they were descended and of which, therefore, they too were inherently a part. Celestial qualities were expressed in the colors of gold and silver and possibly encapsulated in the composition of gilded tumbagas. . . . If we work with Lechtman's suggestion that in Andean thought the surface material or condition of something expressed its inherent essence, then to cover or sheath a wall or object or person in golden color or to entirely cre-

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> ate an object from segments of golden (or gilded tumbaga) sheets was simply to state that the structure, object, or person internally contained, or was composed of, golden (celestial) qualities. In other words, the realm of the Inca nobility, including the nobility themselves and all that surrounded them, was not just *associated* with celestial goldness but was considered as *inherently* "golden" in essence, quality, and concept by virtue of being composed or constructed of goldness. In this interpretive context, then, Incaic golden objects are not viewed simply as golden imitations of nature or as beautiful utilitarian or decorative items but as tangible expressions of the political ideology legitimizing the superiority of the elite of the Inca state.

Of course, the point behind the "technology of essences" is that the essence must be part of the structure of the item in order to be realized and made visible on its surface. Helms has turned the argument around, asserting that the Inca nobility manipulated the ideology behind the technology to validate its claim to inherent godliness. Thus stone walls sheathed with gold or gilt metal or rulers clothed in gold had conferred on them automatically the essential quality that *ought* to have been present internally.

To return to the technology itself: When we step back and look at the two quite extraordinary metal-coloring systems invented by Andean metalworkers—electrochemical replacement and depletion gilding and silvering—it is striking to realize that we are dealing not with the metallurgy of gold or of silver but rather with the metallurgy of copper, the metal that served as the basis for all the major alloys developed in the Andes, including bronze. But unlike its use in some weaponry, in textile manufacturing tools like needles and spindle whorls, and in occasional agricultural implements, copper was transformed when used as the carrier of color.

Most students of early metallurgies consider "true metallurgy" as involving two important achievements: the winning of metals from their

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ores and the alloying of metals. We are used to thinking of copper as the "first" smelted metal and of copper-arsenic as the "first" alloy system. But that is because our data have been taken almost exclusively from the Old World, and even there recent evidence indicates that lead smelting may predate copper smelting by several millennia. From a comparative stance and a fresh look at events in the New World, it seems quite likely that "true metallurgy" in both its aspects—smelting (of copper-silver ores) and alloying (in the production of solders) was achieved in the Andes by the end of the Early Horizon and before the emergence of the Mochica state. But the Mochica put metallurgy on its feet. Copper became the metal on which all further developments in alloying and in smelting were based. In and of itself, copper was highly valued among the Mochica; many of their tools were made of it, and it formed the basis of their alloys. But in this latter role copper was in the service of attitudes about metals—metals of adornment, metals of status, metals of power—that influenced those alloys. They were alloys that altered the properties of copper not so much to make it harder or stronger but to confer on its surfaces a different color—indeed, to transform it so that it actually seemed to be another metal altogether, sometimes metallic silver but primarily metallic gold.

The copper-silver and copper-gold alloy systems developed during the Early Intermediate Period, particularly on the north Peruvian coast, had to be malleable, hard, and capable of retaining the shape they assumed on plastic deformation through hammering. North coast metallurgy was sheet metal oriented, and the alloys it developed had to fulfill the requirements of a sheet metal tradition. Both copper-silver and copper-gold did so admirably. In addition, the Mochica used these alloys to produce metal qualities that were obviously paramount for them, goldenness and silveriness. In this sense, copper, through its alloys, was a vehicle for achieving those desired qualities. It remained a handmaiden to gold and

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to silver, both of which continued to be used in the manufacture of the finest objects.

The interest in surfaces and in the display of gold as the significant visual aspect conveyed by a metal object led to the development not only of *tumbaga* alloys but also of gilding copper by means of external systems as complex as electrochemical deposition. The sophistication of Mochica metallurgy—one is tempted to say of Mochica chemistry—is extraordinary. It gives us insight into what "metalness," at least in part, may have meant to the Mochica, and certainly what resourceful means they used for achieving such qualities.

Ultimately, these attitudes became Andean values indoctrinated by the Inca dynasty. The gold-silver focus within Andean society was, by Inca times, transformed into a state-monopolized industry and one to which the state committed substantial resources of labor and energy, especially in the extraction and processing of the metals. They remained metals of "charisma," concentrated in the hands of the few and redistributed as tokens of royal esteem.

Copper and bronze were also highly valued, but the content of that value is more difficult to assess. The balance between gold and silver, on the one hand, and copper and its alloys, on the other, was never an equal balance, probably because the former always had behind them the weight of religious and political institutions. Copper was nevertheless the vehicle through which the real achievements in Andean metallurgy took place, and it was on the threshold of assuming a much more important role in Andean life when Andean civilization was cut down by Spanish invaders in search of the rich gold and silver deposits of the land.

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